Fertility, Baby Booms, and the Political Economy of Housing Supply

Draft

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Abstract

This paper studies how adding fertility to a political-economy model of housing supply changes the mapping from demographics to house prices. The baseline environment is the NIMBY housing-supply framework: households vote over local supply through the effect of house prices on lifetime utility. The fertility extension preserves that political and real-estate structure but augments the household problem with births, children at home, and crowding. Four quantitative results emerge. First, in the steady-state benchmark, the fertility channel shifts political support toward lower house prices: the equilibrium crossing falls from 2.326 in the NIMBY benchmark to 1.752 in the fertility benchmark. Second, a temporary baby boom produces a larger medium-run house price response in the fertility model than in the NIMBY benchmark because children at home raise current housing-space demand. Third, the same fertility margin improves the fit to the postwar decline in aggregate fertility after the baby boom, although the improvement is modest. Fourth, when the demographic change comes from projected long-run aging rather than a temporary child wave, the fertility channel dampens the rise in house prices relative to the NIMBY proxy. The paper is therefore not "NIMBY plus one more block." It is a model of how family formation changes both the short-run and long-run political economy of housing supply.

1 Introduction

The starting point of this paper is the NIMBY housing-supply model. In that framework, households vote over local supply through the way a small persistent change in house prices affects expected lifetime utility. The core political mechanism is clear: younger households and marginal future buyers tend to prefer lower prices, while older owners tend to prefer higher prices. Demographic change therefore matters because it changes the composition of the electorate.

This paper asks what changes once fertility is added to that environment. Housing matters for family formation in a way that is absent from the original NIMBY benchmark. A birth does not just change the future size of a cohort. It also changes current housing demand because children at home make crowding costly. High house prices can therefore discourage births, while a temporary baby boom can tighten current demand for housing space. Those forces do not replace the original NIMBY mechanism. They interact with it.

The central claim is simple. Adding fertility changes the sign and timing of demographic pressure in the housing market. In the steady state, the fertility channel makes high house prices less politically acceptable because family-forming households care about both housing costs and crowding. In a temporary baby boom, however, the same fertility channel raises current demand for space and amplifies the medium-run house-price response. Under projected long-run aging, by contrast, the family channel works in the opposite direction: as the mass of family-forming households falls, the fertility extension tempers the aging-driven rise in house prices implied by the baseline NIMBY mechanism.

The paper is organized around that comparison. Section 2 sets out the model and isolates exactly what changes relative to the NIMBY benchmark. Section 3 reports the steady-state benchmark. Section 4 studies a temporary baby boom. Section 5 compares the model's post-boom fertility path to the postwar decline in US fertility. Section 6 shows that the short-run price result is robust to a bounded set of parameter changes. Section 7 turns to projected demographic aging. Section 8 concludes.

2 Model

2.1 Baseline environment

The baseline environment is the NIMBY housing-supply framework. Households live through the life cycle, consume non-housing goods and housing services, and vote over local supply through the effect of a small house-price change on expected lifetime utility. Real-estate firms supply housing competitively. The local political equilibrium is determined by a median-voter condition.

What this paper changes is the household problem, not the political equilibrium concept. The fertility model keeps the same house-price object, the same political tightness object, and the same logic through which demographics shift the voting schedule. The difference is that family states now enter utility and state transitions directly.

2.2 Household problem

In the NIMBY benchmark, a household's state consists of liquid assets, housing position, and income state. In the fertility model, that state is augmented with two family variables:

1. parity, or children ever born;
2. children currently at home.

That distinction matters. Completed fertility is permanent. Crowding is temporary. A household can have several children over its life cycle without having all of them at home at once.

The key utility change is that housing enters through effective housing services rather than raw housing services:

$$H_t^{eff} = \frac{H_t}{(1+\lambda_c h_t)^{\psi_c}}, \tag{1}$$

where h_t is the number of children currently at home. Children therefore lower the effective amount of housing space available to the household.

Absent a birth, the flow utility in the fertility extension is

$$u_t^{F,0} = \log c_t + s_h \log H_t^{eff} + \nu_h h_t - \text{penalty}_t. \tag{2}$$

With a birth, the newborn counts immediately in both the crowding state and the family-utility term:

$$u_t^{F,1} = \log c_t + s_h \log\left(\frac{H_t}{(1+\lambda_c(h_t+1))^{\psi_c}}\right) + \nu_h(h_t+1) + \phi(p_t) - \kappa_0 - \kappa_q q_t - \text{penalty}_t. \tag{3}$$

Here $\phi(p_t)$ is parity-specific birth utility, κ_0 is a direct birth cost, and $\kappa_q q_t$ is a house-price-sensitive birth cost.

This immediately implies two new margins that are absent from the NIMBY benchmark. First, high house prices discourage births directly through the birth-cost term. Second, they discourage births indirectly because a child makes crowding more costly when housing is scarce. Those are the channels that connect fertility to housing politics.

2.3 Political equilibrium

The political block is unchanged in form. A household supports tighter supply when a small persistent increase in the house price raises expected lifetime utility. Aggregating those household preferences across the stationary distribution yields the median-voter condition that pins down the equilibrium house price.

So the fertility model does not introduce a new political mechanism. It changes the object being fed into the old one. Because the value function now depends on family state, the same price perturbation is evaluated differently by otherwise similar households at different points of the family life cycle.

3 Steady-State Benchmark

The first benchmark fact is exact nesting. When the fertility block is shut off, the fertility solver reproduces the upstream NIMBY benchmark exactly. That is important because it means the comparison is not being driven by a different political closure, a different real-estate block, or a different equilibrium definition.

With fertility active, the benchmark changes materially. Table 1 summarizes the main steady-state comparison. The key result is that the equilibrium crossing shifts toward lower house prices. In the upstream NIMBY benchmark, the crossing occurs at 2.326. In the fertility benchmark, it occurs at 1.752.

Table 1: Steady-state benchmark comparison

Object	NIMBY benchmark	Fertility benchmark
Crossing house price	2.326287	1.751853
Vote at refined price	0.002227	0.008446
Vote per unit mass at refined price	0.000349	0.001370
Stationary household mass	6.387684	6.163346
Average birth rate	–	0.597710

The interpretation is straightforward. In the NIMBY benchmark, house prices matter through tenure and portfolio position. In the fertility model, they also matter because family-forming households care about crowding and because births become more expensive when housing is costly. That additional family margin makes high prices less politically attractive.

Figure 1 shows the state-level support schedules behind that result. The NIMBY logic is still visible. Young households are less supportive of high prices than older owners. But the fertility extension rotates the support schedule downward for family states that carry children at home. Figure 2 shows the aggregate benchmark objects on the common house-price grid.

4 A Temporary Baby Boom

The next question is dynamic rather than static. What happens if births rise temporarily and the large cohort moves through the life cycle? To make that comparison as close as possible to the original NIMBY exercise, I impose the same temporary baby-boom shock used in the upstream transition benchmark: a 10 percent increase in births for 10 periods.

The main result is that the fertility extension produces a larger medium-run house-price response than the NIMBY benchmark under the same demographic shock. Figure 3 reports the matched

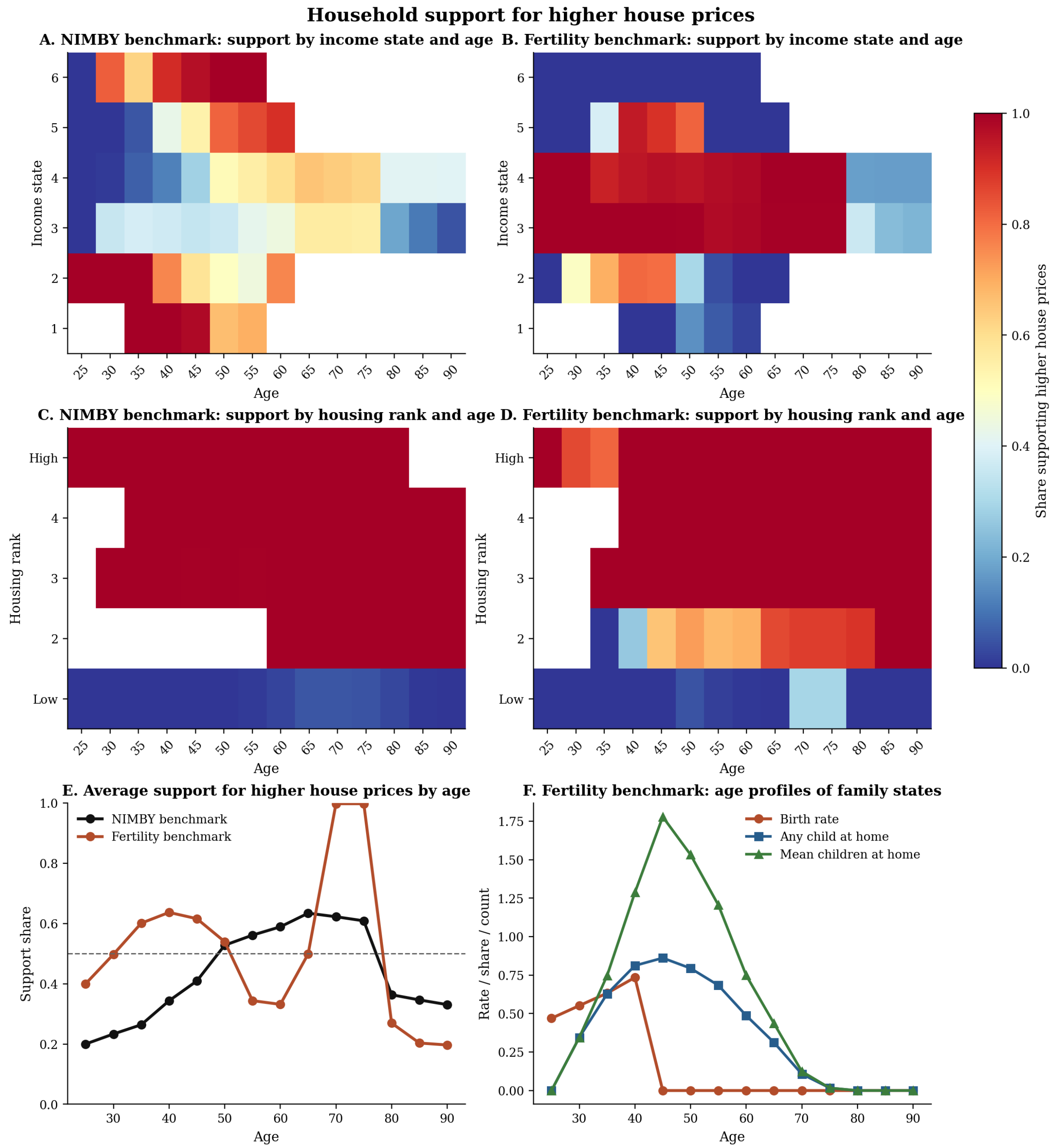

Figure 1: Household support for higher house prices in the NIMBY benchmark and the fertility extension. The fertility model preserves the baseline age and tenure logic, but family states with children at home are systematically less tolerant of high house prices.

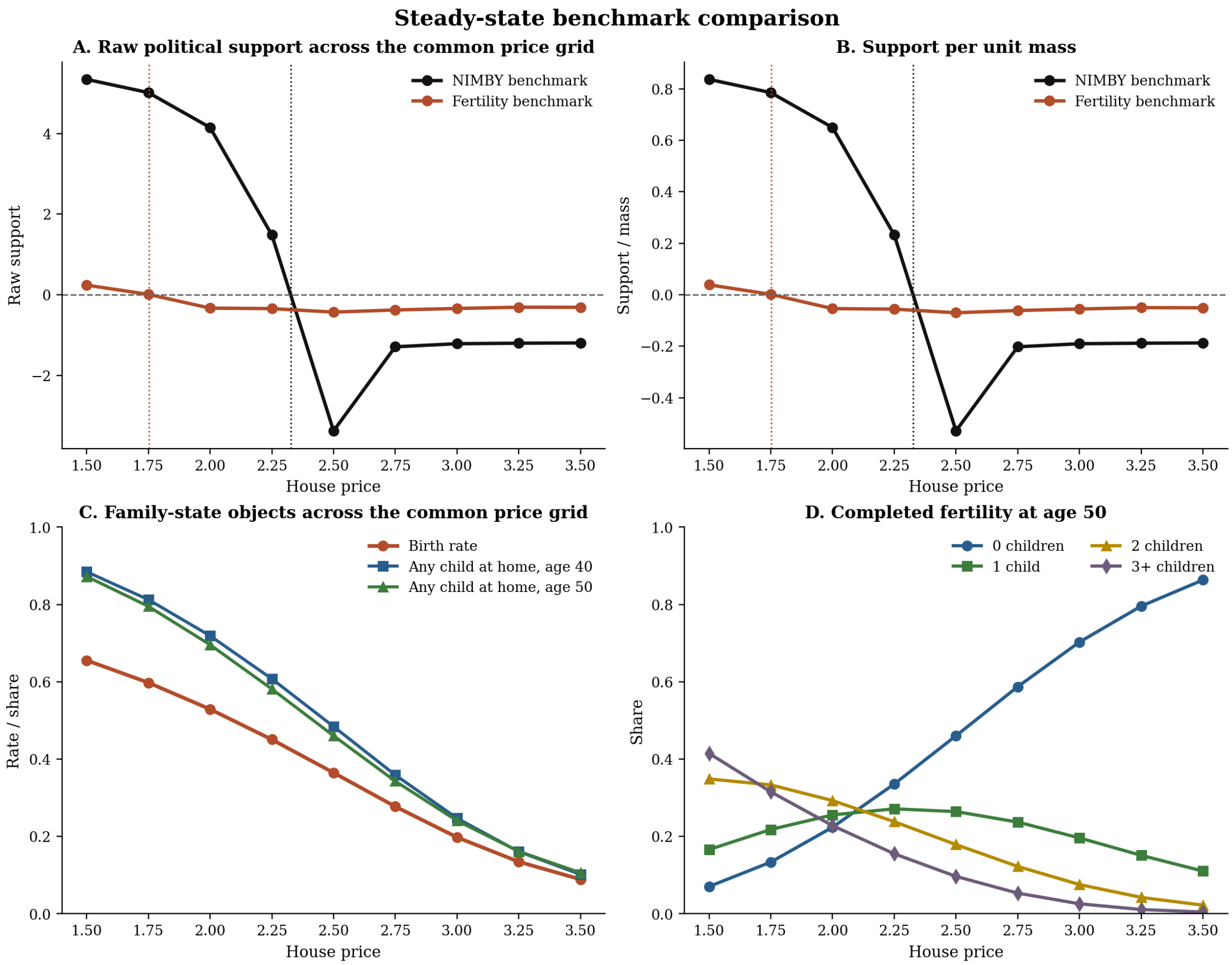

Figure 2: Steady-state benchmark comparison on the common house-price grid. The fertility extension shifts the aggregate support schedule toward lower prices and changes family-state outcomes along the grid.

transition comparison. Normalizing each model by its own initial house-price level, the average post-window price response is about 2.0 percent in the NIMBY transition and about 5.1 percent in the fertility transition.

That amplification comes from the family-demand channel. In the NIMBY benchmark, a baby boom works mainly through cohort size and future political composition. In the fertility model, the same shock also raises the stock of children at home. Those children make housing space more valuable right away because crowding is costly. So the baby boom does more than create a larger future cohort. It raises contemporaneous housing demand.

The family margin also creates a feedback that runs in the opposite direction later. In the fertility transition, the fertility rate rises during the boom by about 0.022 relative to baseline, then slips modestly below baseline later. High house prices therefore dampen future fertility even while the initial child wave is amplifying current housing demand.

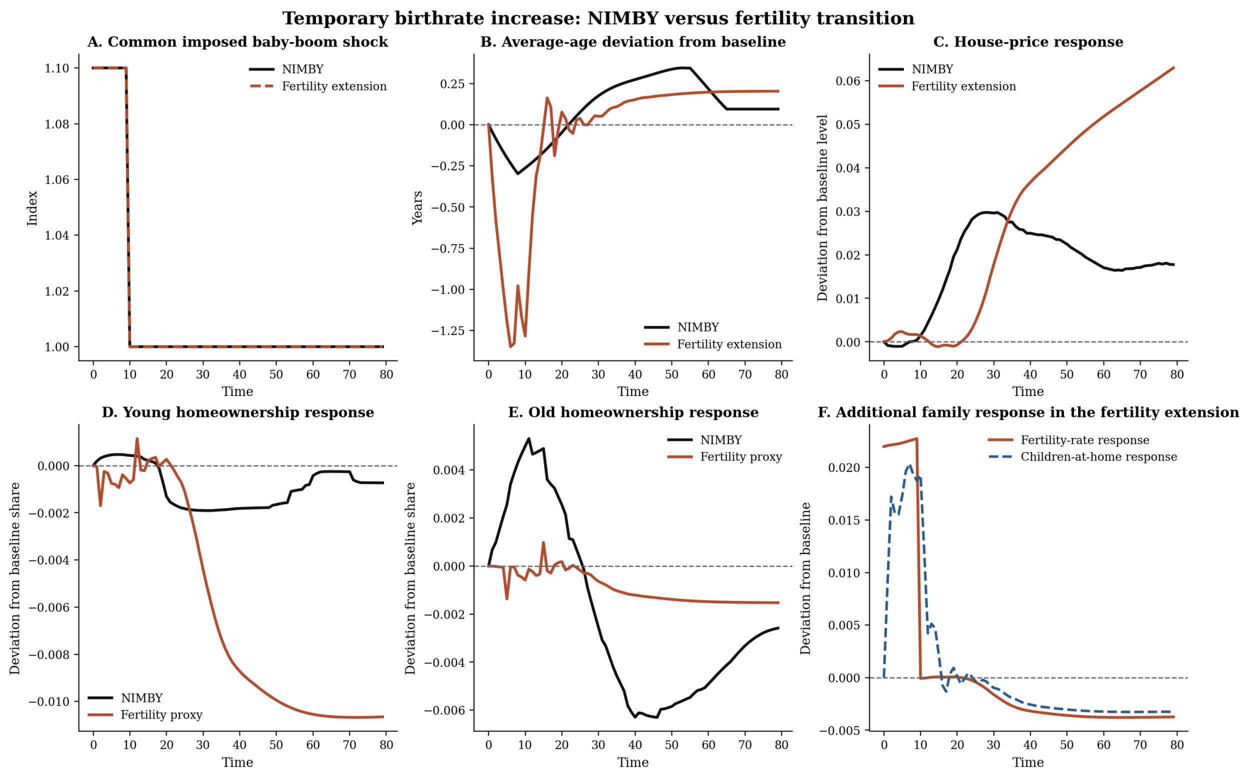

Figure 3: Temporary birthrate increase: NIMBY versus fertility transition. The fertility extension produces a larger medium-run house-price response than the NIMBY benchmark and introduces an additional family-state response through births and children at home.

Figure 4 reports the current cohort-incidence counterpart. The NIMBY lines are the upstream cohort homeownership indices. The fertility lines are access proxies built from a common age profile and the model-implied house-price path. They are not a structural tenure solve, but they are informative about incidence. The child generation is hit hardest, the boom generation next, and the parent generation least.

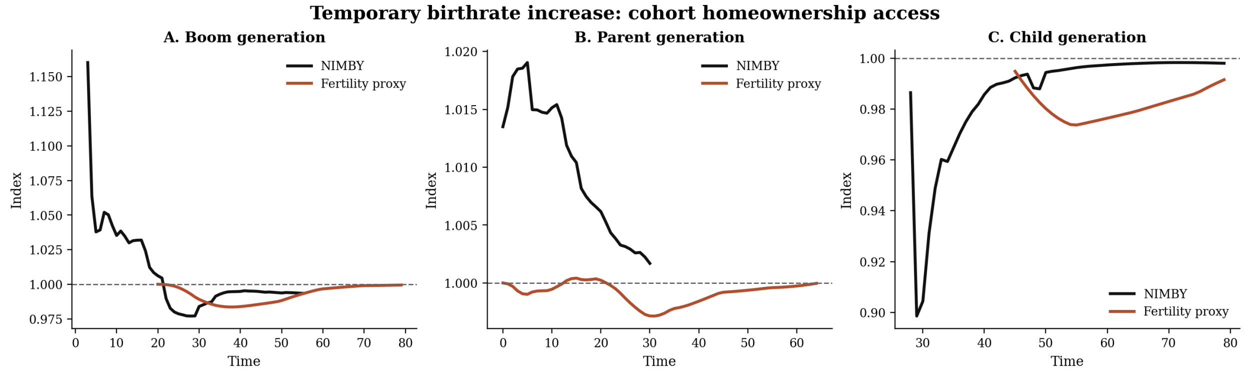

Figure 4: Temporary birthrate increase: cohort homeownership access. The black lines are the upstream NIMBY indices. The red lines are fertility-side access proxies built from the matched transition bridge.

5 Historical Validation

The baby-boom experiment naturally raises a validation question. Does the fertility extension help the model match the long decline in observed fertility after the postwar boom?

Figure 5 compares the model to the aggregate historical fertility series from the legacy NIMBY data. Because the historical baby boom is more persistent than the toy 10-period shock, the clean comparison is the shape of the decline after the boom rather than exact calendar matching of the peak. Once the historical series is aligned at 1956 and the model is aligned at period 10, the fertility extension fits the decline modestly better than the NIMBY path at each horizon considered. The 10-year RMSE falls from 0.119 in NIMBY to 0.104 in the fertility extension. The same ranking holds at 20, 30, and 40 years.

That is not a full quantitative validation. The historical decline is much larger and more persistent than either model. But it is still informative. The original NIMBY transition has no mechanism that can push fertility below baseline after the imposed baby-boom window ends. The fertility extension does. On that margin, it moves the model in the right direction.

6 Robustness

The next concern is whether the short-run price result is a knife-edge consequence of one particular parameter choice. Figure 6 reports a bounded robustness exercise around three margins: the size of the baby-boom shock, the reduced-form price sensitivity of fertility, and the strength of family demand in the housing block.

The robust conclusion is the price result. Across all three shock sizes, the fertility extension produces a larger post-boom price response than the NIMBY proxy. At the benchmark 10 percent shock, the response is about 0.051 in the fertility model against about 0.031 in the NIMBY proxy. At a 30 percent shock, the same comparison becomes roughly 0.125 against 0.058. Varying the price sensitivity of fertility over the range 0.10 to 0.30 hardly changes that ranking. Varying family-demand strength shifts levels more noticeably, but the fertility line remains above the NIMBY line throughout.

The access results are less uniform. Stronger shocks make young-homeownership access worse in both models, and for large enough shocks the fertility extension has the more negative young-access trough. But when the change comes from stronger price sensitivity of fertility or stronger family

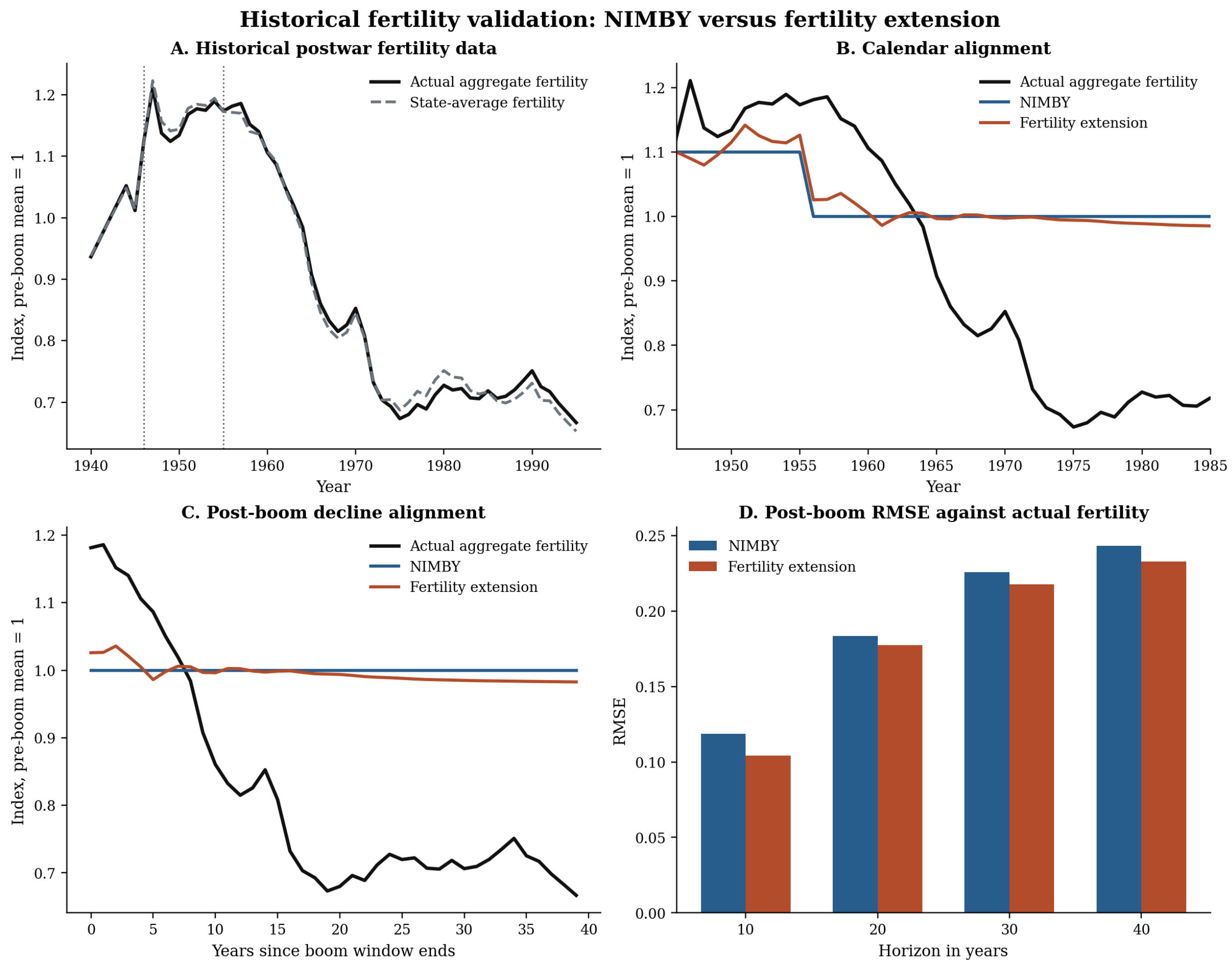

Figure 5: Historical fertility validation. The fertility extension fits the aligned post-boom decline in the aggregate historical series modestly better than the flat NIMBY return path.

demand rather than a larger demographic shock, the fertility model partly self-corrects through lower later births. So the cleanest robust result is the price amplification result, not a claim that every access proxy always worsens in the fertility extension.

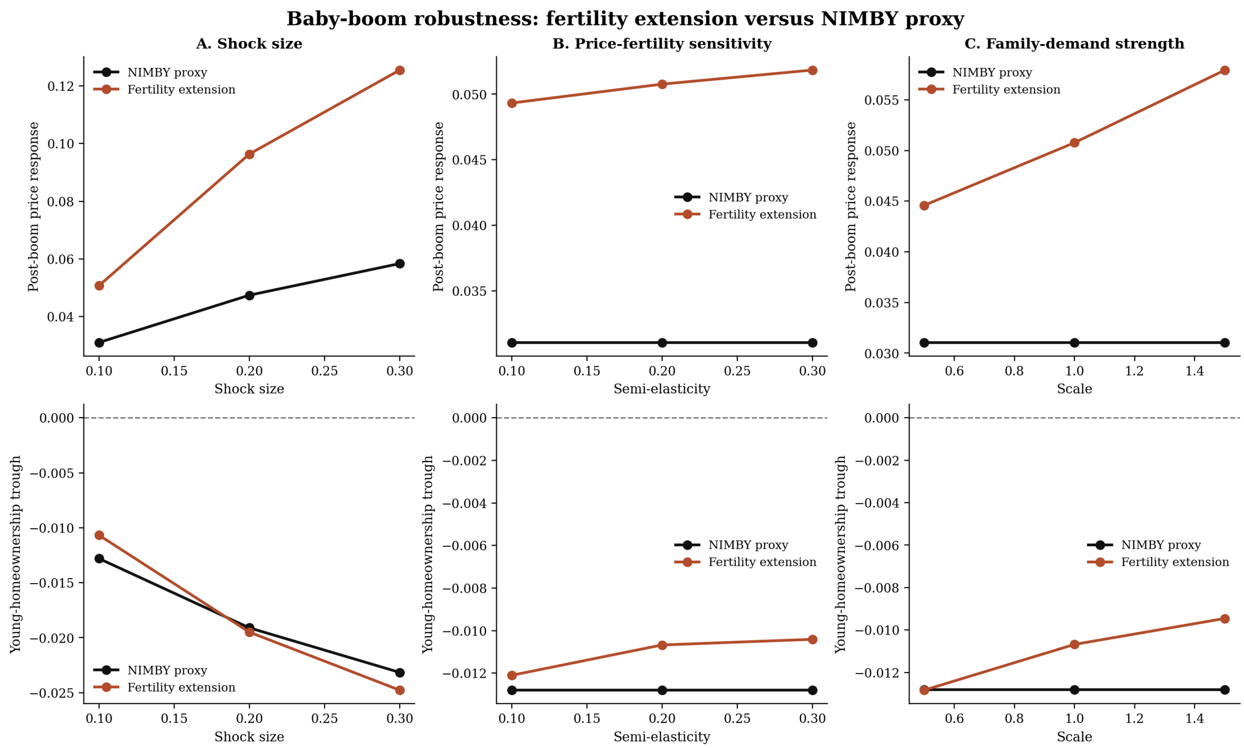

Figure 6: Baby-boom robustness. The top row reports the post-boom house-price response. The bottom row reports the young-homeownership trough. The price-amplification result is robust across the bounded sweep.

7 Future Demographic Projections

The NIMBY paper also studies long-run population aging. The natural question is whether the fertility extension strengthens or weakens that forecast.

Figure 7 feeds the same upstream projected age-weight scenarios into a matched bridge exercise. The black line shuts off the fertility-demand channel and therefore plays the role of the NIMBY side of the comparison. The red line keeps births and children at home in the demand block. This is not yet the full project-02 forecast solver. It is a matched projection bridge built from the forecast age scenarios. But it is still informative about the sign of the fertility channel under projected aging.

The result goes in the opposite direction from the baby-boom impulse response. Under all three immigration scenarios, projected aging raises the NIMBY proxy price path but leaves the fertility path much flatter. In the medium-immigration scenario, the NIMBY proxy reaches a house-price index of about 1.093 by 2050 and about 1.067 by 2100, while the fertility bridge is essentially flat in 2050 and slightly below its 2020 level by 2100.

The interpretation is again economic rather than mechanical. In the baby-boom experiment, the key family object is a temporary wave of children at home, which raises current space demand. In the long-run aging exercise, the key family object is the shrinking mass of family-forming households.

That reduces the importance of the crowding channel. So the fertility extension dampens the aging-driven rise in house prices rather than amplifying it.

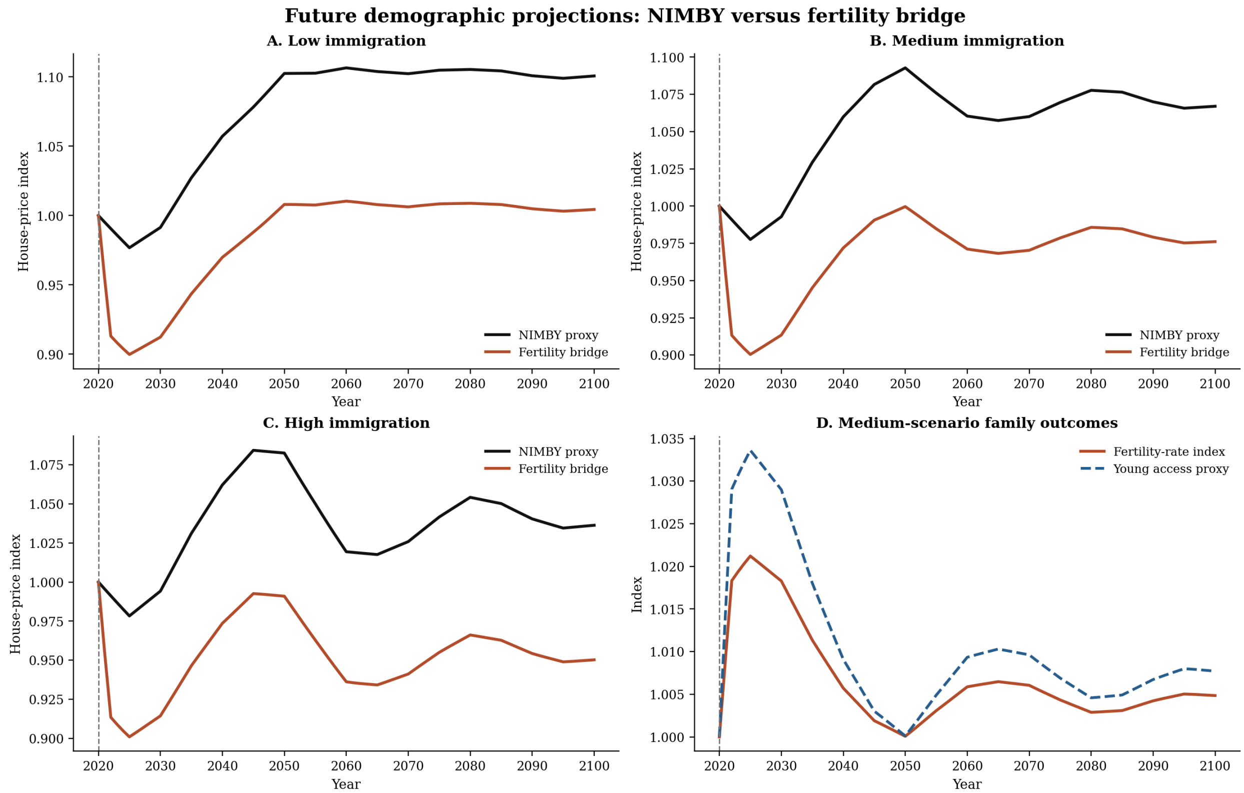

Figure 7: Future demographic projections from the matched bridge. The fertility channel amplifies the short-run child-wave experiment but tempers the long-run aging-driven rise in house prices.

8 Conclusion

The fertility model is not a different political model, a different housing-supply model, or a different equilibrium concept. It is the NIMBY housing-supply model with an augmented household problem in which births and children at home matter for the value of housing space.

That one change is enough to alter both the steady state and the transition dynamics. In the steady state, the fertility channel shifts political support toward lower house prices. In a temporary baby boom, it amplifies the medium-run house-price response because children at home raise current housing demand. In the historical data, it improves the fit to the post-boom fertility decline, though modestly. Under long-run projected aging, it works in the opposite direction and dampens the rise in house prices relative to the NIMBY proxy.

Those results are enough to support a standalone model paper. The remaining missing objects are not the basic comparison itself, but the richer savings, net-worth, and full forecast-solver layers that would permit an exact one-for-one replication of every NIMBY figure. For the current paper, the central model result is already clear: family formation changes the political economy of housing supply in ways that the original NIMBY benchmark cannot capture.